

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-1156
Project Name	EduGenie: Google Gemini Powered learning Assistant
Maximum Marks	5 Marks

Performance Testing
Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust & Postman (API Performance Benchmarking)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	/api/v1/gemini/generate-response (Gemini Assistant Endpoint)
Test Environment	Hosted Cloud Server (AWS EC2 / Node.js Backend)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load - Concurrent student Q&A chat requests	50	300	Response time < 2s, 0% failure rate
2	Peak Load - Document summary generation requests	150	600	Response time < 5s, error rate < 1%
3	Spike Testing - Sudden influx of students opening quiz generator	300	120	System handles burst without crashing
4	Endurance Testing - Sustained interactive study session load	100	1800	Stable memory and CPU usage below 80%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.45 seconds	Pass	Optimal
2	Response Time (Max)	< 5 seconds	3.82 seconds	Pass	Under peak load
3	Throughput (Req/sec)	> 50 req/sec	68 req/sec	Pass	Sufficient bandwidth
4	Error Rate	< 1%	0.25%	Pass	Minor timeouts resolved

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
5	CPU Utilization	< 80%	62%	Pass	Healthy load balance
6	Memory Utilization	< 80%	58%	Pass	No memory leaks detected

Step 4: Observations & Analysis

Key Findings:

The EduGenie API platform maintained stable performance across normal and peak load tests. The Google Gemini API integration responded within acceptable thresholds, with an average latency of 1.45s for standard Q&A queries.

Bottlenecks Identified:

Slight latency increases were observed during high-volume document summary requests (USN-7) when payload size exceeded 5MB.

Optimization Steps Taken:

Implemented response caching for repeated student queries, optimized payload parsing, and configured connection pooling for asynchronous Gemini API requests.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Locust Load Testing Summary Report Graph Placeholder - EduGenie API Benchmarks]